



[Psychology of Memory]

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Al-Ameed College of Medicine
[Academic year]

Learning outcomes

By the end of this session, students should be able to:



- **Define memory** and explain its three main processes: encoding, storage, and retrieval.
- **Describe the main models of memory** and their basic stages.
- **Differentiate between sensory, short-term (working), and long-term memory.**
- **Differentiate between declarative and procedural memory**, including semantic and episodic memory.
- **Describe the main brain areas involved in memory** and factors that can affect memory.
- **Recognize common memory disorders** and describe simple strategies to improve memory.

Memory

Encoding

Storing

Retrieving



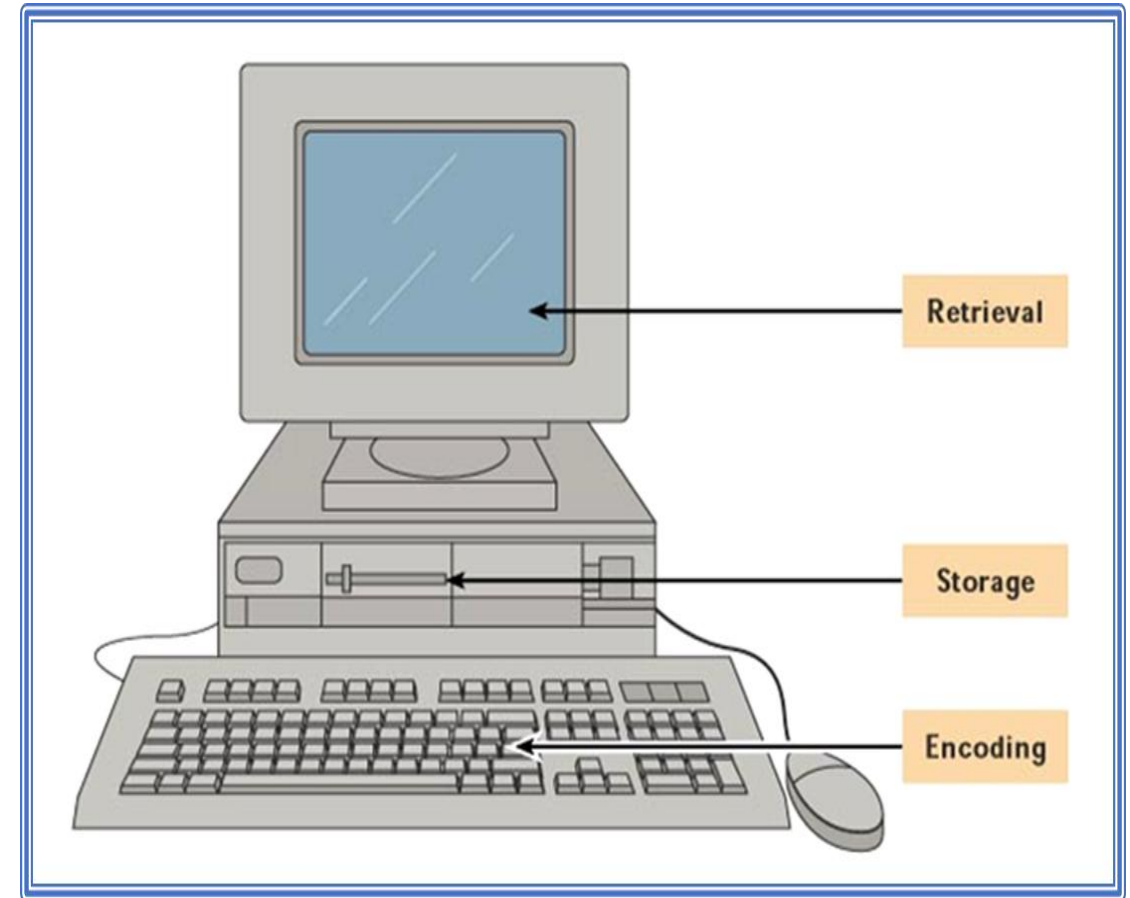
Definitions of memory

[What are the three processes?

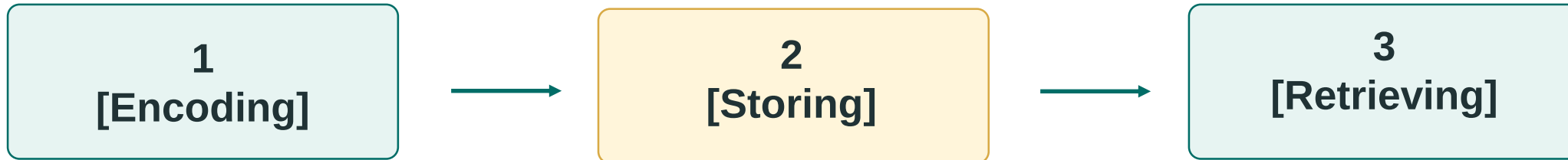
Key points

Memory is the ability to retain information over time through three processes: **encoding (forming), storing, and retrieving.**

Memories are not copies but representations of the world that vary in accuracy and are subject to error and bias.



Mechanism / process



1-Encoding

Encoding refers to making mental representations of information so that it can be placed into our memories.

2 Storing

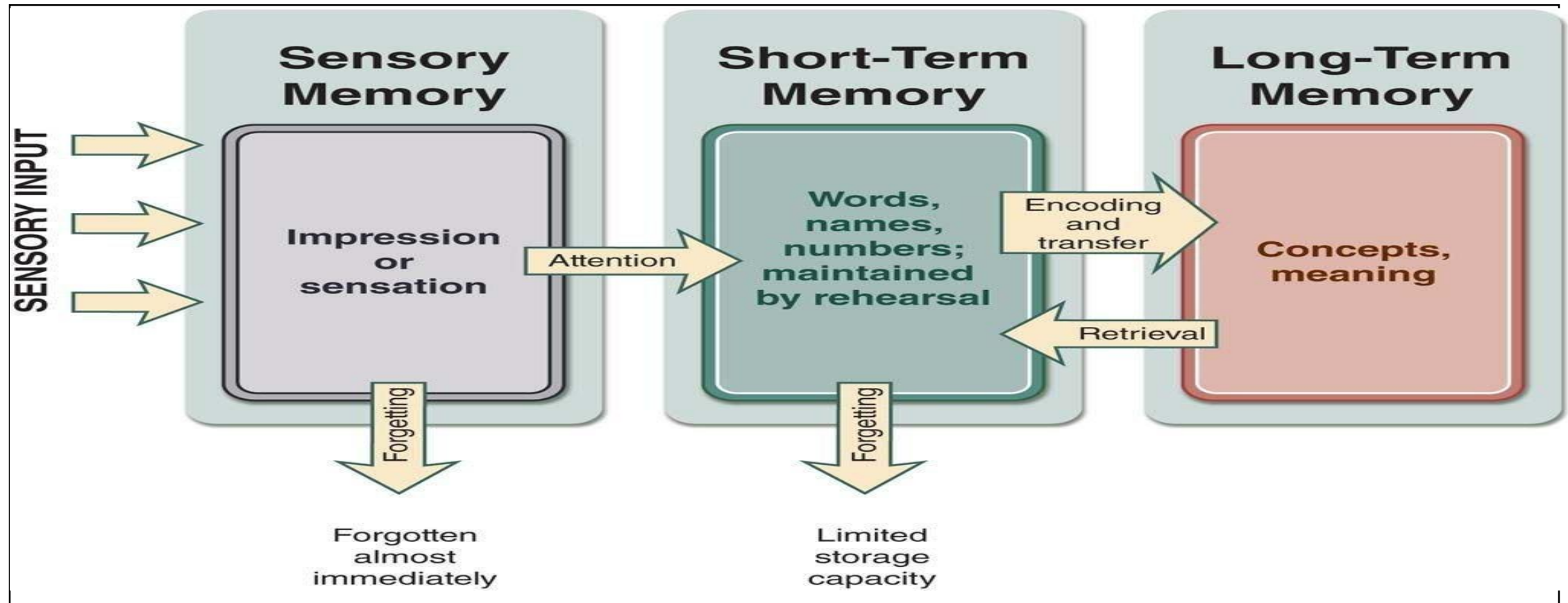
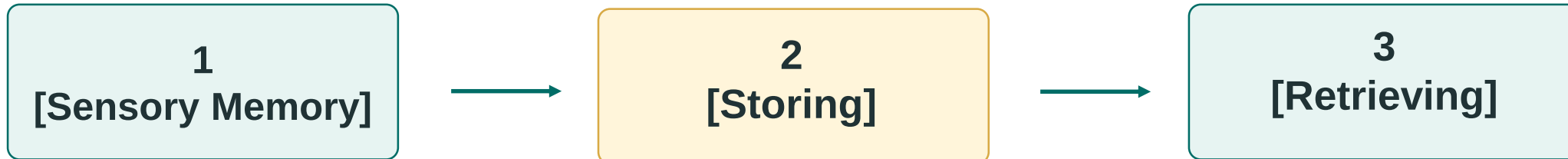
Storing is the process of placing encoded information into relatively permanent mental storage for later recall.

New information that is stored by making associations with old or familiar information is much easier to remember, or retrieve.

3 Retrieving

Retrieving is the process of getting or recalling information that has been placed into short-term or long-term storage.

Three Types of Memory



1-Sensory Memory

Key points

1-Sensory Memory

Imagine you are walking through a **busy shopping mall**. You see many things and hear different sounds, such as a **guitarist playing music**.

All these sights and sounds briefly enter your **sensory memory**.

Sensory memory is the first stage of memory. It receives information from the environment and keeps it for a **very short time—from an instant to a few seconds**.

For example,

the guitarist's music may remain in your **auditory sensory memory for about 1–2 seconds**.

- If you **do not pay attention**, the information disappears.
- If you **pay attention**, the information moves from **sensory memory → short-term memory**.



Sensory Memory: Recording

Do you have mental video recorder?

Your brain has a mental video-audio recorder that automatically receives and holds incoming sensory information for only seconds or less.

It is brief period provides just enough time for you to decide whether some particular incoming sensory information is important or interesting and therefore demands your further attention.

We'll examine two different kinds of sensory memory: visual sensory memory, called iconic memory, and auditory sensory memory, called echoic memory.

Sensory Memory: Recording

Iconic Memory (Visual Sensory Memory)

- A form of sensory memory that briefly holds visual information.
- Lasts for about 0.25 second (a quarter of a second) or slightly longer.
- When you shift your attention, the information disappears.
- The word “iconic” means image.

Example: Blinking



We do not experience darkness during a blink because the visual information is briefly stored in iconic memory.

Echoic Memory (Auditory Sensory Memory)

- A form of sensory memory that holds auditory information.
- Lasts for about 1–2 seconds.
- Allows you to briefly “play back” sounds that you thought you did not hear.
- Helps you recognize that sequences of sounds form words.
- The length of echoic memory increases as children grow into adults.

Example: Friend asking a question



Even after you say “What did you say?”, you can still recall your friend’s exact words because they are still in echoic memory (up to 2 seconds).

Comparison of Iconic and Echoic Memory

Feature	Iconic Memory (Visual)	Echoic Memory (Auditory)
Type of information	Visual (images)	Auditory (sounds)
Duration	About 0.25 second or slightly longer	About 1–2 seconds
Example	Seeing a scene during an eye blink	Briefly remembering a spoken sentence
Function	Maintains visual information briefly	Holds sounds long enough to recognize speech and play back what was just heard

Example: When you blink, your eyes are completely closed for a short moment, but you do not experience darkness. This is because the visual information is briefly maintained in iconic memory.

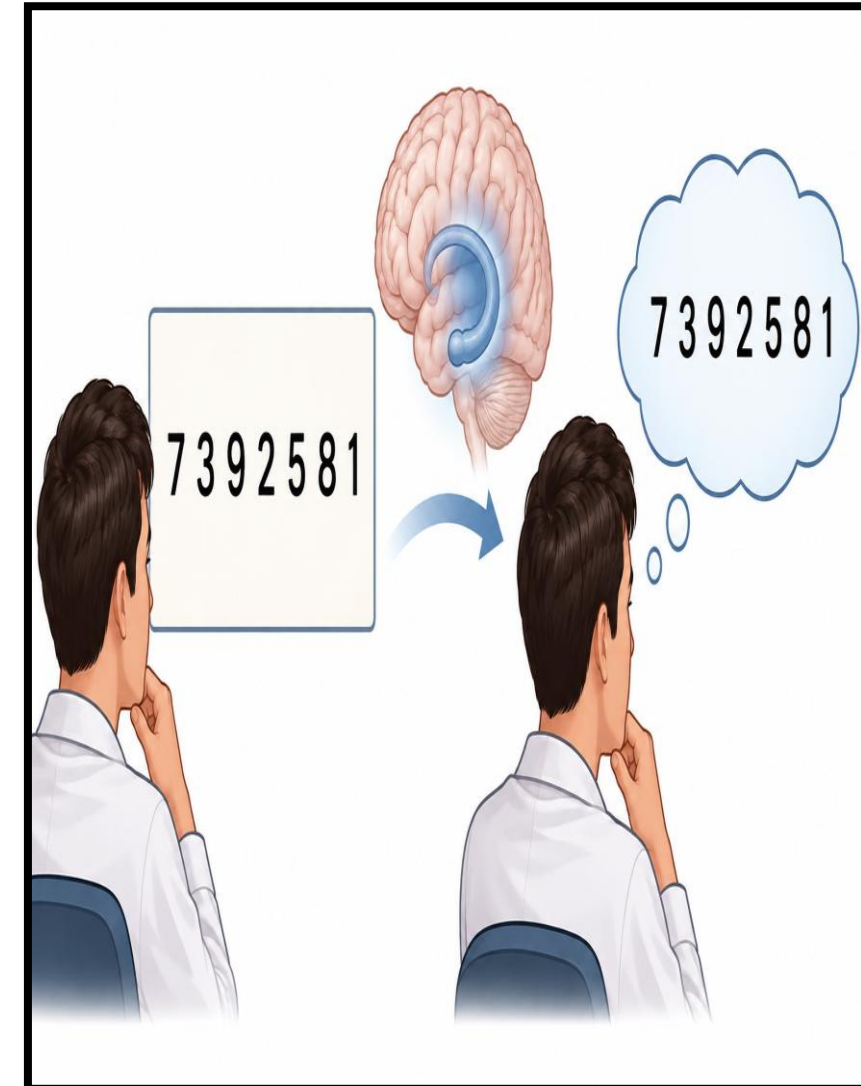
Sperling's Experiment, participants briefly saw 12 letters arranged in 3 rows.

The letters were shown for only 50 milliseconds.-

Immediate recall: about 9 letters-

0.5-second delay : about 6 letters-

1-second delay: about 4 letters



2-Short-Term Memory

Key points

2-Short-Term Memory

Because the guitarist's music sounds interesting, you **pay attention** to it.

This attention transfers the information from **sensory memory** to **short-term memory**. working or short term memory has the two characteristics of limited duration and limited capacity.

Short-term memory (working memory) holds a limited amount of information—about **7 items**—for a short time, usually **2–30 seconds**.

For example, you may remember a few notes of the guitarist's song for a short time. If you **hum the song or think about it**, the information may stay in short-term memory longer.

If you do not continue to use the information, it will disappear. If you **rehearse or process it**, it can be transferred to **long-term memory**.



2-Short-Term Memory

Key points

Working or short term memory has the two characteristics of limited duration and limited capacity.

❑ Limited Duration

The new telephone number that you looked up will remain in short-term memory for a brief time, usually from 2 to 30 seconds, and then disappear. However, you can keep information longer in short-term memory by using maintenance rehearsal.

Maintenance rehearsal refers to the practice of intentionally repeating or rehearsing information so that it remains longer in short-term memory.

❑ Limited Capacity

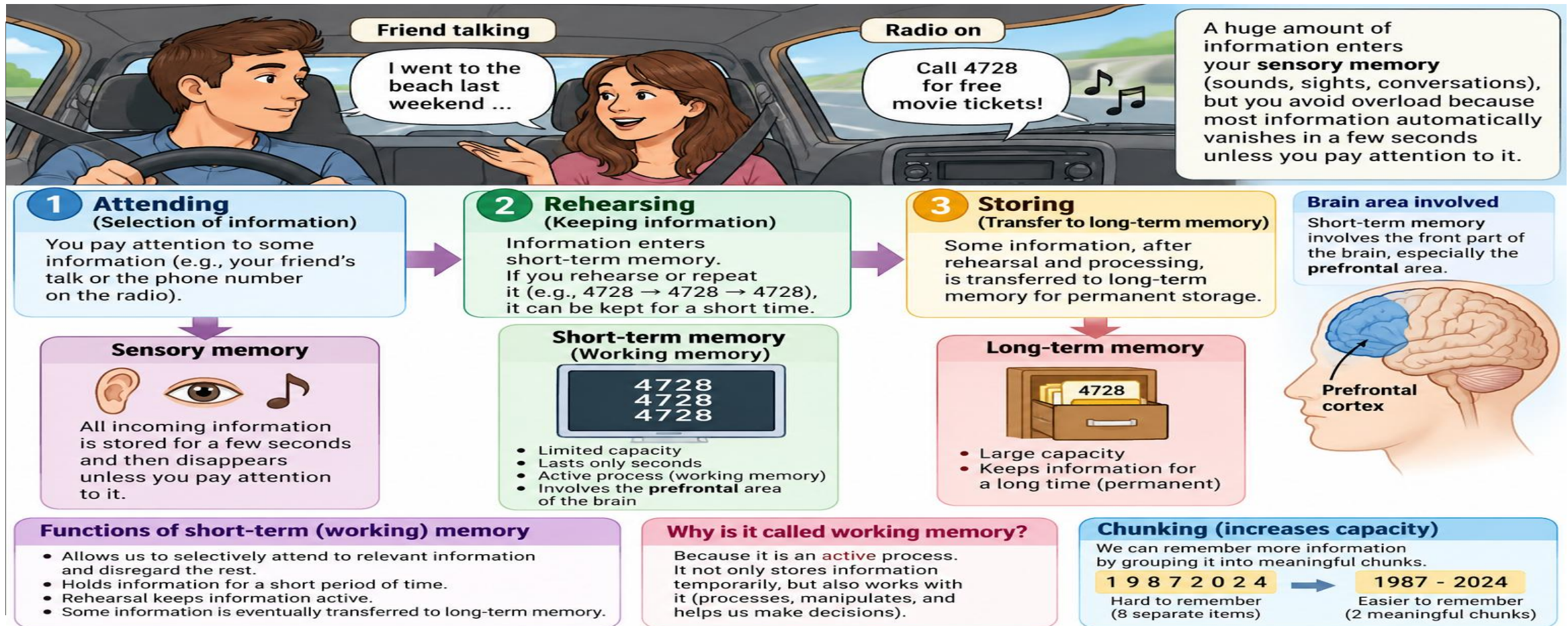
One of the main reasons information disappears from short-term memory is interference.

Interference results when new information enters short-term memory and overwrites or pushes out information that is already there.

For example, students make few errors when they are asked to repeat seven or eight digits, make some errors with a list of eight or nine digits, and make many errors when they repeat a list that is longer than nine digits.

2-Short-Term Memory

Key points



3-Long-Term Memory

Key points

3-Long-Term Memory

Long-term memory refers to the process of storing almost unlimited amounts of information over long periods of time.

If you continue to think about the guitarist's music—for example, **whistling along** or thinking about why the music sounds familiar—the information may move from **short-term memory to long-term memory**.

Long-term memory stores a very large amount of information for a **long period of time**.

For example, you may have stored many **songs, words, faces, and conversations** in your long-term memory. This information can be retrieved when you need it, although sometimes you may have difficulty remembering something that you know you have learned.



3-Long-Term Memory

Key points

Long-term memory can be divided into **two main types**:

1. Declarative memory (الذاكرة التصريحية)

2. Procedural (non declarative) memory

1. Declarative Memory : is memory for **facts and events** that we can consciously remember and describe.

For example:

- **Which bird cannot fly?** → Penguin
- **What did you eat for breakfast?** → This is a personal event.

There are **two types of declarative memory**:

A. Semantic Memory : is memory for **facts, knowledge, words, concepts, definitions, and rules**.

Examples:

- Knowing that a **penguin cannot fly** , Knowing the definition of **depression** , Remembering the meaning of a medical term , Remembering facts learned in class

3-Long-Term Memory

Key points

B. Episodic Memory: is memory for **personal experiences and specific events**.

Examples:

- What you ate for breakfast , Your first day at medical college, A trip you took last year, A specific conversation you had yesterday , Your experience at a particular restaurant

3-Long-Term Memory

Key points

2. Procedural Memory (الذاكرة الإجرائية) (Nondeclarative Memory) :is memory for **skills and learned behaviors**.

It helps us remember **how to do something**, even when we cannot explain exactly how we learned or perform the skill. We cannot recall or retrieve procedural memories.

Examples:

- Riding a bicycle ,Playing tennis ,Driving a car ,Typing ,Reading ,Playing a musical instrument

For example, you may be able to **ride a bicycle after many years without practice**, but you may not be able to explain every movement your muscles make.

Procedural memory also includes some emotional responses

Some emotional behaviors can be learned through **classical conditioning**.

For example:

- A person may develop a **fear of spiders** after a frightening experience.
- The person may feel fear automatically but may not be able to explain exactly how the fear was learned.



Core concept 5 : Compare & contrast

[Compare two related ideas clearly]

A

Declarative Memory

Facts and events
Conscious memory
Can be recalled and described
Knowing what
Example: knowing the definition of depression

B

Procedural Memory

Skills and behaviors
Mostly unconscious
Difficult to consciously recall
Knowing how
Example: knowing how to drive

Long-Term Memory

Declarative Memory (Explicit memory)

Memory for **facts and events** that we can consciously remember and describe.

Semantic Memory Facts and knowledge

- Knowledge of facts, concepts, words, definitions and rules.
- Most of what you learn in classes.

Example:

- Which bird cannot fly?
- Definition of depression.
- Meaning of a medical term.



Episodic Memory

Personal events and experiences

- Knowledge of specific events and personal experiences.
- Involves when, where and what happened.

Example:

- What did you eat for breakfast?
- Your first day at medical school.
- A trip you took last year.
- A specific conversation.



Procedural Memory (Nondeclarative memory)

Memory for **skills and learned behaviors**. Also includes some emotional behaviors.

- Stored mostly unconsciously.
- We can perform the skill but cannot easily explain how we do it.
- Includes motor skills, some cognitive skills (e.g., reading), and emotional responses learned through conditioning.



Example:

- Riding a bicycle.
- Playing tennis.
- Driving a car.
- Playing a musical instrument.
- Fear of spiders (learned through classical conditioning).



Key Differences

Feature	Declarative Memory	Procedural Memory
Content	Facts and events	Skills and behaviors
Awareness	Conscious	Mostly unconscious
Recall	Can be recalled and described	Difficult to recall or describe
Type of knowledge	Knowing what	Knowing how
Examples	Definition, personal events	Riding a bicycle, playing tennis

H.M. Example



- After brain surgery, H.M. could not form new declarative memories.
- He could not remember new facts (semantic).
- He could not remember new events (episodic).
- However, he could learn and improve motor skills (procedural), e.g., mirror-drawing.

Key point: H.M. could learn a skill without remembering that he had learned it.

Quick check: MCQ

Which sequence correctly represents the three main memory processes?

- A. Encoding → storing → retrieving
- B. Storing → encoding → retrieving
- C. Retrieving → storing → encoding
- D. Encoding → retrieving → storing
- E. Storing → retrieving → encoding

Quick check: MCQ

Memories are best described as:

- A. permanent recordings without error
- B. exact copies of the world
- C. representations of the world
- D. purely emotional responses
- E. sensory impressions only

Quick check: MCQ

Retrieving means:

- A. getting or recalling stored information
- B. forming mental representations
- C. placing information into permanent storage
- D. receiving sensory input
- E. overwriting old information

Quick check: MCQ

Attention helps information move from sensory memory to:

- A. semantic memory directly
- B. long-term memory directly
- C. procedural memory
- D. episodic memory directly
- E. short-term memory

Quick check: MCQ

Sensory memory receives information mainly from:

- A. the environment
- B. long-term storage
- C. the hippocampus only
- D. previous dreams
- E. motor cortex only

Quick check: MCQ

Attention helps information move from sensory memory to:

- A. procedural memory
- B. B. long-term memory directly
- C. short-term memory
- D. episodic memory directly
- E. semantic memory directly

Quick check: MCQ

Sensory memory automatically receives and holds incoming information for:

- A. only seconds or less
- B. several minutes
- C. hours
- D. days
- E. Months

Quick check: MCQ

Visual sensory memory is called:

- A. echoic memory
- B. iconic memory
- C. semantic memory
- D. episodic memory
- E. procedural memory



Quick check: MCQ

A person remembering the last few seconds of a spoken sentence is using:

- A. echoic memory
- B. iconic memory
- C. semantic memory
- D. episodic memory
- E. procedural memory

Quick check: MCQ

Rehearsing information can:

- A. eliminate attention
- B. B. immediately erase it
- C. prevent encoding
- D. turn it into sensory memory
- E. keep it in short-term memory longer



Quick check: MCQ

1. Declarative memory is memory for:

- A. facts and events that can be consciously remembered and described
- B. motor skills only
- C. conditioned reflexes only
- D. sensory impressions only
- E. unconscious habits only



Quick check: MCQ

Remembering what you ate for breakfast is an example of:

- A. episodic memory
- B. semantic memory
- C. procedural memory
- D. iconic memory
- E. echoic memory

Quick check: MCQ

Procedural memory is memory for:

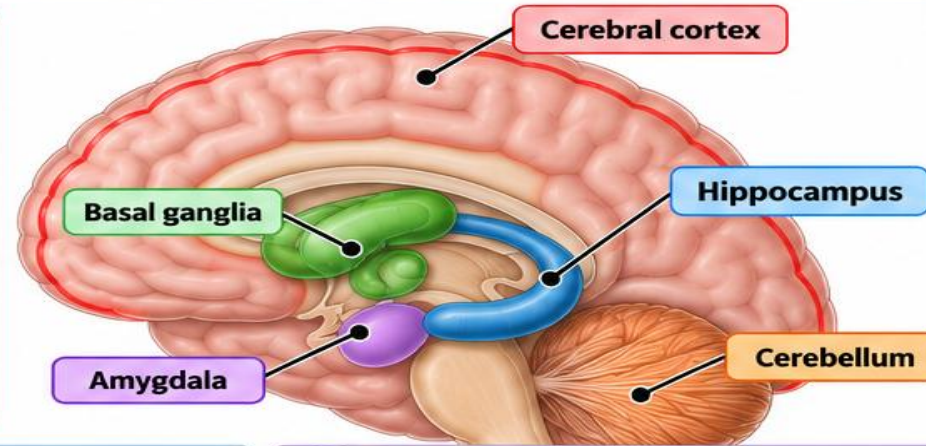
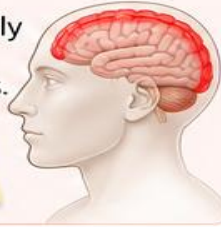
- A. facts and definitions
- B. skills and learned behaviors
- C. personal events only
- D. sensory impressions
- E. telephone numbers

Brain Areas Involved in Memory

1 Cortex: Short-Term Memory

- The cortex helps hold information in **short-term memory**.
- Example:** remembering a phone number long enough to dial it.
- If the cortex is intact, a person may have short-term memory and relatively normal conversations even if they cannot form new long-term memories.
- Short-term memory is temporary and limited.

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2 Cortex: Long-Term Memories

- Long-term memories are stored in many areas throughout the **cerebral cortex**.
- This includes words, songs, facts, and events.
- Information learned before brain damage may still be remembered because it is already stored in the cortex.



3 Hippocampus: Transferring Declarative Memories

- Acts like a **"Save" button**.
- Transfers declarative information from short-term memory into long-term memory.
- This includes words, facts, and personal events.
- Damage to the hippocampus → cannot form new declarative memories.
- Procedural skills can still be learned.

Short-term memory
(words, facts, events)



Long-term memory
(declarative)

4 Amygdala: Emotional Memories

- Adds **emotional meaning** to memories.
- Especially important for fear and threatening situations.
- Helps recognize emotional facial expressions.
- Damage to the amygdala may leave the memory intact but reduce its emotional impact.



5 Procedural Memory

- Stored in **basal ganglia, cerebellum** and **motor cortical areas**.
- Involves motor skills, habits and some cognitive skills (e.g., learning to read).
- Not dependent on the hippocampus.**
- Example:** riding a bicycle, playing tennis, typing, tying shoes.



Basal ganglia
(skills and habits)



Cerebellum
(motor coordination)



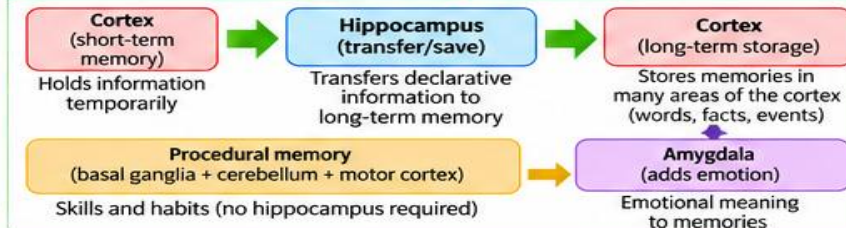
Motor cortex
(learned motor skills)



Summary of Brain Areas in Memory

Brain area	Main function	Type of memory	Examples
Cerebral cortex	Holds short-term information and stores long-term memories	Short-term and long-term (declarative)	Words, songs, facts, events
Hippocampus	Transfers declarative information from short-term to long-term memory	Declarative	New words, facts, personal events
Amygdala	Adds emotional content to memories	Emotional memory	Fear, threat, positive and negative emotions
Basal ganglia	Stores procedural information (skills and habits)	Procedural (nondeclarative)	Habits, motor skills (e.g., riding a bicycle)
Cerebellum	Stores procedural information (motor coordination)	Procedural (nondeclarative)	Balance, coordination (e.g., playing tennis)
Motor cortex	Involved in storage and execution of learned motor patterns	Procedural (nondeclarative)	Typing, playing an instrument, skilled movements

Simple Memory Model



Common Causes of Memory Problems

1. Age

As people get older, **memory may decline slightly**. This is often called **age-associated memory impairment**.

Examples:

- Forgetting names, Misplacing objects, Occasionally forgetting information

2. Stress and Anxiety: High levels of **stress and anxiety** can interfere with: Concentration, Learning, Memory retrieval

Stress hormones can also affect the **hippocampus**, which is important for memory formation.

3. Lack of Sleep

Not getting enough sleep can impair: Concentration, Learning, Memory

Poor sleep → poorer concentration → poorer memory.

4. Medical Conditions Some medical conditions can cause memory problems, such as:

- **Thyroid disorders, Vitamin B12 deficiency, Infections. Head injuries, Alzheimer's disease, Other forms of dementia**

5. Medications and Substance Use

Some medications may affect **memory and cognition**, particularly some drugs used for:

- Anxiety, Sleep disorders, Depression

Alcohol and other substances can also impair memory.

Psychopathology of memory

1. Amnesia: Amnesia, involves the partial or complete loss of memory. It can be caused by physical injuries, medical conditions, psychological trauma, or certain psychiatric disorders.

Retrograde amnesia: old memories lost

Anterograde amnesia: new memories lost

Transient Global Amnesia: Transient global amnesia is a temporary form of amnesia that usually lasts for a few hours

2. Memory Intrusions: Memory intrusions refer to the **involuntary and unwanted recall of distressing or traumatic memories**. They are commonly associated with conditions like **PTSD**.

3. Memory Distortions: Memory distortions involve **inaccuracies or alterations in memory recall**. Memory distortions are commonly observed in conditions like **depression, anxiety disorders, and schizophrenia**.

Disorders of Memory

1 Amnesia

- Amnesia involves the **partial or complete loss of memory**.
- It can be caused by physical injuries, medical conditions, psychological trauma, or certain psychiatric disorders.



Retrograde Amnesia

Loss of **old (past)** memories

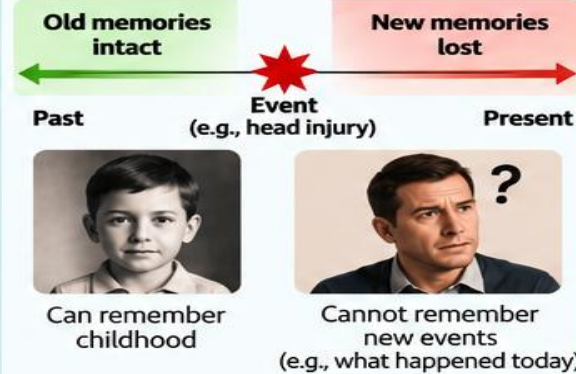
- Inability to recall memories that occurred before the amnestic event.
- New memories can still be learned and remembered.



Anterograde Amnesia

Loss of **new** memories

- Inability to form or remember new memories after the amnestic event.
- Old memories (before the event) are usually preserved.



Transient Global Amnesia (TGA)

Temporary form of amnesia

- Sudden onset of anterograde amnesia, often with some retrograde amnesia for recent events.
- Usually lasts for a few hours (typically 1–24 hours).
- The person is awake, alert and can perform complex tasks, but repeatedly asks the same questions and cannot form new memories during the episode.
- Full recovery – no lasting memory problems.



2 Memory Intrusions

- Memory intrusions refer to the **involuntary and unwanted recall of distressing or traumatic memories**.
- They are commonly associated with conditions like **PTSD**.

Examples:

- Recurrent intrusive memories
- Flashbacks
- Nightmares



3 Memory Distortions

- Memory distortions involve **inaccuracies or alterations in memory recall**.
- They are commonly observed in conditions like **depression, anxiety disorders, and schizophrenia**.

Examples:

- False memories (remembering events that did not occur)
- Exaggerated or biased memories
- Distorted recall of details



Quick check: MCQ

Which brain area is described as transferring declarative information into long-term memory?

- A. Hippocampus
- B. Amygdala
- C. Cerebellum
- D. Basal ganglia
- E. Motor cortex

Quick check: MCQ

Mild memory decline associated with normal aging is called:

- A. transient global amnesia
- B. anterograde amnesia
- C. retrograde amnesia
- D. age-associated memory impairment
- E. memory intrusion



Quick check: MCQ

Anterograde amnesia involves loss of:

- A. emotional memory only
- B. old memories
- C. sensory memory only
- D. motor skills only
- E. new memories